01.1 supplement · Sets, measurable maps and Bertrand's paradox

Data Analytics for AI · Executive M.Tech · IIT Jammu
Module 01 · Fundamentals of Probability and Random Signal Theory
Supplement to 01.1 · Introduction to probability through problems

01.1 main materials

- **Notes:** Main lecture notes (PDF)
- **Deck:** Worked student deck (PDF)

Supplementary materials

- **Notes:** This rigorous supplement (PDF)
- **Deck:** Probability before Kolmogorov (PDF)
- **Deck:** Bertrand's paradox (PDF)
- **Interactive demonstration:** Sampling maps and chord length

This supplement gives the formal definitions, parametrizations and derivations supporting the main lecture. Its details are intended for supplementary study alongside the corresponding lecture topics. For the chord calculations, R is the fixed circle radius, T the normalized midpoint distance, and L the chord length, as in the Bertrand deck. The Billing indicator in the main notes remains X_B.

The main lecture motivates these constructions with questions about a hypothetical six-ticket routing audit, before introducing the triplet and axioms through finite calculations. The proofs below do not depend on that application; they state the general hypotheses needed for the results.

Reference: Krishna Jagannathan, *Probability Foundations for Electrical Engineers*, 9 July 2025, Chapters 1, 4 and 11, particularly Definitions 68–70 for measurable functions, random variables and their laws. The calculations of the sampling maps are written out below.

1. Outcomes, events and mutual exclusivity

A sample space is required to satisfy two conditions. It must be **exhaustive**, so that at least one of its elements occurs in any realization of the experiment, and its elements must be **mutually exclusive**, so that at most one of them occurs. Together these state that exactly one $\omega \in \Omega$ occurs each time the experiment is performed.

Distinct outcomes correspond to disjoint singleton subsets. When these singletons belong to the event collection, they are singleton events:

$$\{\omega_1\} \cap \{\omega_2\} = \varnothing \quad \text{whenever } \omega_1 \neq \omega_2.$$

For countable Ω the singletons therefore form a countable disjoint partition, $\Omega = \bigcup_{\omega \in \Omega} \{\omega\}$ (a disjoint union).

Outcomes are mutually exclusive; events need not be. For a single roll of a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$, and the outcomes 2 and 4 cannot both occur. The events

$$A = \{2, 4, 6\} \text{ (even)}, \qquad B = \{4, 5, 6\} \text{ (greater than 3)}$$

can both occur, since $A \cap B = \{4, 6\} \neq \varnothing$.

Consequences for calculation

The two cases are handled by different rules, and the distinction above is what separates them.

Because singletons are disjoint, countable additivity applies directly to them, and for a countable event A whose singletons are measurable

$$P(A) = \sum_{\omega \in A} P(\{\omega\}).$$

For two events no such rule is available, since additivity requires disjointness. The correction term is the shared part:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Writing $P(A \cup B) = P(A) + P(B)$ for the die events above would count the outcomes 4 and 6 twice.

Two errors that this distinction prevents

A list of events is not a sample space. For one roll of a die, the pair “even” and “greater than 3” is not a sample space: both can occur in a single roll, so the second condition fails. They are events in the finer space $\{1, \ldots, 6\}$.

A valid sample space need not have equally likely elements. For two independent fair tosses, the list “a head on the first toss”, “a tail then a head”, “two tails” is exhaustive and mutually exclusive, so it is a legitimate sample space. Its members carry probabilities $1/2$, $1/4$ and $1/4$: they group the equally likely ordered outcomes as $\{HH, HT\}$, $\{TH\}$ and $\{TT\}$. This calculation also appears in the interpretations deck. Mutual exclusivity is a requirement on the sample space; it says nothing about the probabilities assigned to its elements, and counting the members is valid only when they happen to be equally probable.

The same reading in the Bertrand construction

The sampling procedures in Section 3 produce a geometric chord almost surely. After excluding the stated null cases, we may describe their outputs on the common space $\mathcal{C}_*$ of nondegenerate, nondiameter chords. In this output description, a chord is an elementary outcome. The three rotationally invariant models use the same chord space and event collection but different probability measures. Their original sampling outcomes can instead be angles, distances or points; Section 3 distinguishes these descriptions.

The event $E = \{c : L(c) > \sqrt{3}\ R\}$ is not an outcome. It is a set containing uncountably many chords, and it is assigned a probability as a set.

This last point also marks the limit of the summation rule above. In a continuous model each individual outcome has probability zero, so

$$\sum_{c \in E} P(\{c\}) = 0 \neq P(E),$$

and the probability of E cannot be recovered by adding the probabilities of the outcomes it contains. Countable additivity is additivity over countably many disjoint events, and E is not a countable union of its singletons. The rule $P(A) = \sum_{\omega \in A} P(\{\omega\})$ applies to countable events with measurable singletons; it does not extend to an uncountable event by countable additivity.

Event collections and the probability triplet

The main notes use all subsets as events for finite experiments. Here are the precise requirements for a general triplet $(\Omega, \mathcal{F}, P)$.

Definitions: power set and sigma-algebra

The power set $\mathcal{P}(\Omega) = 2^{\Omega} = \{E: E \subseteq \Omega\}$ contains every subset, including $\varnothing$ and Ω. For finite $|\Omega| = n$, it has 2^n elements, since including or excluding each outcome gives two choices. An event collection $\mathcal{F}$ is a subset of $\mathcal{P}(\Omega)$, so its members are themselves subsets of Ω.

The event collection $\mathcal{F}$ must be a **sigma-algebra** on Ω. Its requirements are:

1. **Contains the sample space:** $\Omega \in \mathcal{F}$.
2. **Closed under complements:** if $E \in \mathcal{F}$, then $E^c = \Omega \backslash E \in \mathcal{F}$.
3. **Closed under countable unions:** if $E_1, E_2, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} E_i \in \mathcal{F}$.

These conditions ensure that "not E" and "at least one of $E_1, E_2, \ldots$" are events whenever the component sets are events. Since this example has a finite sample space, taking $\mathcal{F} = 2^{\Omega}$, the power set, satisfies all three requirements.

No disjointness assumption is needed for closure under unions. It is needed when applying countable additivity of probability. Closure also gives $\varnothing \in \mathcal{F}$, countable intersections by De Morgan's law, and differences $E \backslash F = E \cap F^c$.

The probability measure P is nonnegative, has $P(\Omega) = 1$, and is countably additive on pairwise disjoint events. These conditions and the event-collection requirements together define a probability space.

In the six-ticket audit, $\Omega = \{t_1, \ldots, t_6\}$ and $B = \{t_1, t_2, t_5\}$.

A smaller event collection

Suppose a dashboard records only whether the selected ticket requires Billing. It distinguishes $B = \{t_1, t_2, t_5\}$ from $B^c = \{t_3, t_4, t_6\}$, but does not identify individual tickets. The events available in this description are

$$\mathcal{G} = \{\varnothing, B, B^c, \Omega\}.$$

This is a sigma-algebra: complements exchange the pairs, and any union of its members is again one of the four members. Restrict the uniform-ID measure to $\mathcal{G}$; the four probabilities are $0, 1/2, 1/2, 1$ respectively. The singleton $\{t_1\}$ is a subset of Ω but is not an event in $\mathcal{G}$. The full audit model still uses $\mathcal{F} = 2^{\Omega}$; the smaller model illustrates why the event collection must be specified.

A continuous triplet on the unit interval

Suppose a simulation selects a number uniformly from the unit interval. What is the probability that the selected value lies between $1/4$ and $3/4$? Uniformity means that an interval's probability is its length, so the answer should be $3/4 - 1/4 = 1/2$. To make this a probability model, we must specify more than intervals: "outside this interval" and "in at least one of these intervals" must also be events.

Take $\Omega_u = [0, 1]$. A sample point $u \in [0, 1]$ is a number; an event is a suitable **set of numbers**. We use the Borel σ-algebra

$$\mathcal{F}_u = \mathcal{B}([0, 1]) = \sigma\big(\{(a, b) \cap [0, 1]: a, b \in \mathbb{R},\ a < b\}\big).$$

Here $\sigma(\mathcal{A})$ means the smallest σ-algebra containing every member of a family $\mathcal{A}$. Equivalently, it is the intersection of all σ-algebras on $[0, 1]$ that contain that family. Such σ-algebras exist because the power set $2^{[0,1]}$ is one of them. The intervals intersected with $[0, 1]$ generate all the relatively open sets of the unit interval.

A **Borel set** is one member of $\mathcal{B}([0, 1])$. The **Borel σ-algebra** is the collection of all those members. For example,

$$[1/4, 3/4] \in \mathcal{B}([0, 1]), \qquad [1/4, 3/4] \subseteq [0, 1].$$

The first statement says it is a measurable event; the second says its elements are possible outcomes.

What the Borel σ-algebra contains

It contains $\varnothing$, $[0,1]$, every relatively open or closed subset, every interval within $[0,1]$ (including half-open intervals), and every singleton $\{u\}$. It is closed under complements **relative to** $[0,1]$, countable unions and countable intersections. The intersection property follows from De Morgan's law:

$$\bigcap_{n=1}^{\infty} E_n = [0,1] \setminus \bigcup_{n=1}^{\infty} \left([0,1] \setminus E_n\right).$$

For example, each set $[0,1] \cap (u - 1/n, u + 1/n)$ is relatively open, and

$$\{u\} = \bigcap_{n=1}^{\infty} \left([0,1] \cap (u - 1/n, u + 1/n)\right).$$

Thus singletons are Borel. Every countable set of points in $[0,1]$ is then Borel, being a countable union of singletons. In particular, $\mathbb{Q} \cap [0,1]$ is Borel, as is its complement, the irrational points in $[0,1]$.

These examples do not exhaust the collection: The collection includes sets formed through repeated countable unions, intersections and complements, beyond the examples listed. Closure under **countable** operations does not imply closure under arbitrary uncountable unions. The Borel σ-algebra is smaller than $2^{[0,1]}$; not every subset is Borel. Section 5 below constructs a set that is not Lebesgue measurable. Since every Borel set is Lebesgue measurable, that set is not Borel either.

Assigning probabilities and checking the triplet

Let λ be Lebesgue measure, the standard measure that assigns each interval its length, and restrict it to Borel subsets of $[0,1]$:

$$P_u(E) = \lambda(E), \qquad E \in \mathcal{B}([0,1]).$$

No scaling is needed because $\lambda([0,1]) = 1$. Our triplet is

$$\left([0,1], \mathcal{B}([0,1]), P_u\right).$$

To verify it, check the event collection and the measure separately.

- **Event collection.** Every σ-algebra in the defining intersection contains $[0,1]$ and is closed under relative complements and countable unions. Their intersection inherits these properties, so $\mathcal{B}([0,1])$ is a σ-algebra.
- **Nonnegativity.** Lebesgue measure is nonnegative, so $P_u(E) \geq 0$ for each Borel event E.
- **Normalization.** $P_u([0,1]) = \lambda([0,1]) = 1$.
- **Countable additivity.** For pairwise disjoint Borel events $E_1, E_2, \ldots$, the countable additivity of Lebesgue measure gives

  $$P_u\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} P_u(E_n).$$

Here we use the standard theorem that Lebesgue measure exists and is countably additive on Borel sets. Checking a few intervals alone would not establish that theorem. Restricting this measure to the unit interval verifies all the probability axioms, and $P_u(E) \leq 1$ follows from $E \subseteq [0,1]$.

Worked events in the triplet

Borel event E	Probability $P_u(E)$	Reason
$[1/4, 3/4]$	$1/2$	Interval length
$[0, 1/4) \cup (3/4, 1]$	$1/2$	Complement of the preceding event
$\{1/2\}$	0	A singleton has zero length
$\mathbb{Q} \cap [0, 1]$	0	Countably many disjoint singletons of measure zero
$[0, 1] \backslash \mathbb{Q}$	1	Complement of the rational points

The rational points are dense in $[0, 1]$, but still have probability zero in this model. The measure is length, not the number of points or how closely they are spaced.

For an explicit countable-additivity calculation, let

$$I_n = (2^{-(n+1)}, 2^{-n}], \qquad n = 0, 1, 2, \ldots.$$

These Borel intervals are pairwise disjoint and have union $(0, 1]$. Hence

$$P_u((0, 1]) = \sum_{n=0}^{\infty} 2^{-(n+1)} = 1.$$

Adding the zero-probability singleton $\{0\}$ gives $P_u([0, 1]) = 1$. This example illustrates countable additivity. It does not license an uncountable sum of singleton probabilities.

2. Random variables, inverse images and probability laws

On a probability space $(\Omega, \mathcal{F}, P)$, a real-valued **random variable** is a measurable function

$$X: \Omega \to \mathbb{R}.$$

Measurability means that $X^{-1}(B) \in \mathcal{F}$ for every Borel set $B \in \mathcal{B}(\mathbb{R})$. The Borel σ-algebra is generated by open intervals and includes the intervals, singleton sets and their countable combinations used here. For a real-valued function, it is sufficient to check $\{\omega: X(\omega) \leq x\} \in \mathcal{F}$ for every real x.

The distinction between a random variable and a set function

A random variable is a function **on outcomes**, not a real-valued set function on events. It induces the following map between collections of sets:

$$X^{-1}: \mathcal{B}(\mathbb{R}) \to \mathcal{F}, \qquad B \mapsto \{\omega: X(\omega) \in B\}.$$

The probability measure and the distribution of X are real-valued **set functions**:

$$P: \mathcal{F} \to [0, 1], \qquad P_X: \mathcal{B}(\mathbb{R}) \to [0, 1],$$

$$P_X(B) = P\left(X^{-1}(B)\right) = P(X \in B).$$

Thus a numerical question about X is first converted to a set of outcomes, and that event is assigned a probability. This is the set-based construction used in the Bertrand calculation below. It follows the distinction in Jagannathan, Chapter 11, Definitions 68–70.

Independence used in the examples

Events E, F are independent when $P(E \cap F) = P(E)P(F)$. Random variables U, V are independent when this identity holds for the events $\{U \in A\}$ and $\{V \in B\}$ for every pair of Borel sets A, B. For variables with a joint density on a product domain, $f_{U,V}(u, v) = f_U(u) f_V(v)$ almost everywhere implies independence: integrating over $A \times B$ separates into the product of the two marginal integrals. Independent fair tosses below stipulate the product measure, giving each ordered pair mass $(1/2)(1/2) = 1/4$.

Worked example: the head count

For two independent fair tosses, each outcome in $\Omega = \{HH, HT, TH, TT\}$ has probability $1/4$, and $\mathcal{F} = 2^{\Omega}$. Define $X(HH) = 2$, $X(HT) = X(TH) = 1$, and $X(TT) = 0$. Then

$P_X(\{1\}) = P(\{HT, TH\}) = 1/2,$

$P_X([1, 2]) = P(\{HH, HT, TH\}) = 3/4.$

The distribution assigns probabilities $1/4, 1/2, 1/4$ to $0, 1, 2$. It is obtained through inverse images, not by assuming that the three numerical values are equally likely.

Worked example: an indicator

For an event $A \in \mathcal{F}$, its indicator is the random variable

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Its inverse images satisfy $\mathbf{1}_A^{-1}(\{1\}) = A$ and $\mathbf{1}_A^{-1}(\{0\}) = A^c$. Consequently, $P(\mathbf{1}_A = 1) = P(A)$.

For the parity model in the event-collection lab, take $\mathcal{G} = \{\varnothing, \{2, 4, 6\}, \{1, 3, 5\}, \Omega\}$. The indicator of an even face is measurable. The numerical face-value function is not measurable relative to that σ-algebra, since the inverse image of $\{6\}$ is the unavailable subset $\{6\}$.

Measurability and the probability law

Return to the six-ticket audit of Section 1, with $B = \{t_1, t_2, t_5\}$, $X_B = \mathbf{1}_B$, and $\mathcal{G} = \{\varnothing, B, B^c, \Omega\}$.

To calculate $P(X_B \in D)$, the inverse image $X_B^{-1}(D)$ must be an event. For example, $X_B^{-1}((1/2, 3/2)) = B \in \mathcal{F}$, giving probability $1/2$. This requirement motivates the formal definition.

As with the unit interval in Section 1, now on $\mathbb{R}$, the **Borel σ-algebra** $\mathcal{B}(\mathbb{R})$ is the smallest sigma-algebra containing all open intervals. It includes intervals, singletons and their countable unions, intersections and complements. A function $X\colon \Omega \to \mathbb{R}$ is **measurable**, and hence a real-valued random variable on $(\Omega, \mathcal{F}, P)$, if

$X^{-1}(D) \in \mathcal{F} \quad \text{for every } D \in \mathcal{B}(\mathbb{R}).$

Its **law** or **distribution** is the probability measure

$P_X(D) = P(X^{-1}(D)), \qquad D \in \mathcal{B}(\mathbb{R}).$

In the full finite model every function is measurable because all subsets are events. For X_B, the inverse image is one of $\varnothing, B, B^c, \Omega$ according to whether D contains neither of 0 and 1, only 1, only 0, or both. Thus X_B is measurable even for the smaller collection $\mathcal{G}$. Its law assigns mass $1/2$ each to 0 and 1. By contrast, the ID-code function $J(t_i) = i$ is not measurable for $\mathcal{G}$: $J^{-1}(\{1\}) = \{t_1\}$ is unavailable in that collection. It is measurable for 2^{Ω}.

3. Bertrand's paradox: sampling maps and induced distributions

This section develops the Bertrand question posed in Section 3 of the main notes. What is the probability that a randomly generated chord is longer than the side of an inscribed equilateral triangle? We first calculate three answers to this same plain-English question. Only then do we use the Kolmogorov triplet to identify precisely which probability problem each calculation answers.

The paradox: one question, three answers

For the unit circle, a chord at distance r from the centre has length

$L = 2\sqrt{1-r^2}$.

Thus $L > \sqrt{3}$ exactly when $r < 1/2$, or equivalently when the chord intersects the interior of the concentric disk of radius $1/2$.

Consider three plausible readings of "choose a chord at random."

1. **Choose its angle uniformly at a fixed endpoint.** Fix $a = (1, 0)$ and measure an angle $\theta \in (0, \pi)$ from the upward tangent towards the disk. The resulting chord has length $2\sin\theta$. It is longer than $\sqrt{3}$ when $\pi/3 < \theta < 2\pi/3$, an interval occupying one third of the available angular range. The answer is $1/3$.
2. **Choose its distance from the centre uniformly.** Fix the midpoint direction and choose its distance r uniformly between 0 and 1. Draw the chord perpendicular to that radius. The favorable distances satisfy $r < 1/2$, half the available range. The answer is $1/2$.
3. **Choose its midpoint uniformly in disk area.** Draw the chord perpendicular to the radius through the chosen midpoint. It is long enough exactly when the midpoint lies inside the concentric disk of radius $1/2$. That disk occupies one quarter of the total area. The answer is $1/4$. The centre and boundary are zero-area exceptions handled below.

We have asked the same question in words and obtained

$$\frac{1}{3}, \qquad \frac{1}{2}, \qquad \frac{1}{4}.$$

Each calculation is correct for its construction. The phrase "a random chord" leaves the sampling rule unspecified: uniform angle, distance and area are different assumptions. We now use the Kolmogorov triplet to resolve the apparent contradiction.

Making the experiments precise with the Kolmogorov triplet

A probability model is a triplet $(\Omega, \mathcal{F}, P)$: Ω specifies the outcomes, $\mathcal{F}$ specifies the measurable events, and P assigns their probabilities. We now identify these three components for each calculation. We distinguish the common geometric question from the event that represents it in each sampling space.

Experiment I: a uniform angle at a fixed endpoint. Fix the endpoint $a = (1, 0)$. From the upward tangent at a, measure $\theta \in (0, \pi)$ towards the interior of the disk. The ray has direction $(-\sin\theta, \cos\theta)$; its second intersection with the circle defines the chord. Substituting $a + \ell(-\sin\theta, \cos\theta)$ into the circle equation gives $\ell(\ell - 2\sin\theta) = 0$, so its length is $L = 2\sin\theta$.

Choose θ uniformly. The triplet and measure are

$$\Omega_1 = (0, \pi), \qquad \mathcal{F}_1 = \mathcal{B}((0, \pi)), \qquad P_1(B) = \frac{1}{\pi}\int_B d\theta \quad (B \in \mathcal{F}_1).$$

Here $\mathcal{B}$ is the Borel σ-algebra, generated by the open subsets of the stated domain. The favorable event and probability are

$$E_1 = (\pi/3, 2\pi/3), \qquad P_1(E_1) = \frac{2\pi/3 - \pi/3}{\pi} = \frac{1}{3}.$$

The endpoints $\theta = 0, \pi$ would give zero-length chords and have been excluded. The diameter at $\theta = \pi/2$ has probability zero and will also be omitted when we introduce the common chord chart.

Experiment II: a uniform distance with a fixed direction. Choose $r \in (0,1)$ uniformly, set the midpoint to $(r,0)$, and draw the vertical chord through it. Specifying the midpoint direction makes the chord unique; distance alone determines only its length.

$$\Omega_2 = (0,1), \qquad \mathcal{F}_2 = \mathcal{B}((0,1)), \qquad P_2(B) = \int_B dr \quad (B \in \mathcal{F}_2).$$

The favorable event and probability are

$$E_2 = (0,1/2), \qquad P_2(E_2) = \int_0^{1/2} dr = \frac{1}{2}.$$

Experiment III: a uniform midpoint in area. Choose a point $M = (x,y)$ uniformly in the disk and draw the chord perpendicular to OM through M. To make this construction unique and nondegenerate everywhere on its domain, exclude the centre and boundary, both of which have area zero. Put

$$D^* = \{(x,y) \colon 0 < x^2 + y^2 < 1\}.$$

Then

$$\Omega_3 = D^*, \qquad \mathcal{F}_3 = \mathcal{B}(D^*), \qquad P_3(B) = \frac{1}{\pi} \iint_B dx\, dy \quad (B \in \mathcal{F}_3).$$

The favorable event is the punctured inner disk, so

$$E_3 = \{(x,y) \colon 0 < x^2 + y^2 < 1/4\}, \qquad P_3(E_3) = \frac{\pi (1/2)^2}{\pi} = \frac{1}{4}.$$

Deleting the centre does not change either area. Using the closed disk with an explicit convention for the exceptional cases gives the same probabilities.

Experiment	Sample space	Measure on Borel events	Favorable event	Probability
Fixed endpoint, uniform angle	$(0,\pi)$	$d\theta/\pi$	$\pi/3 < \theta < 2\pi/3$	$1/3$
Fixed direction, uniform distance	$(0,1)$	dr	$r < 1/2$	$1/2$
Uniform midpoint area	D^*	$dx\, dy/\pi$	$x^2 + y^2 < 1/4$	$1/4$

The notation for the measures in the table means integration over an event, as defined above. **With the probability triplets specified, the apparent contradiction is resolved:** the three answers belong to three different probability models.

Advanced: measures on a common chord space

The triplets above complete the resolution. The remaining calculations study sampling maps and induced measures on geometric chords. They are not needed to resolve the paradox.

We consider three rotationally invariant models independently:

- **Model A:** choose two independent uniform endpoints on the circumference.
- **Model B:** choose a uniform radial distance and an independent uniform midpoint direction.
- **Model C:** choose a midpoint uniformly in disk area.

Their connection to Experiments I–III is Exercise 8. In particular, equal chord-length distributions need not mean equal measures on geometric chords. A common sample space does not require identical supports.

The endpoint, distance and area constructions are also discussed in Larson and Odoni, *Urban Operations Research*, §3.3.2. We now specify the common measurable space and derive each model's sampling law.

The event and a common chord space

Fix a circle of radius $R > 0$. The side of an inscribed equilateral triangle has length $\sqrt{3}\ R$. For a chord c, define the real-valued random variables

$$D(c) = \text{distance from the centre to } c, \qquad T(c) = D(c)/R, \qquad L(c) = \text{length of } c.$$

The geometry gives

$$L(c) = 2R\sqrt{1 - T(c)^2}, \qquad E = \{c\colon L(c) > \sqrt{3}\ R\} = \{c\colon T(c) < 1/2\}.$$

We compare the probability of this **same event** under three probability measures on a common chord space.

For a one-to-one parametrization, let $\mathcal{C}_*$ be the set of unordered, nondegenerate chords that are not diameters. Coincident-endpoint chords and diameters have probability zero under all three rules below. Excluding them makes the inverse formulas unambiguous without changing any calculated probability. They may instead be retained in an enlarged space with total probability zero.

A chord parametrization and its inverse

Let $0 < t < 1$ and $0 \le \phi < 2\pi$. Write

$$n(\phi) = (\cos\phi, \sin\phi), \qquad n_\perp(\phi) = (-\sin\phi, \cos\phi).$$

Define $C(\phi, t)$ to be the chord with unordered endpoints

$$z_\pm = Rt\, n(\phi)\ \pm\ R\sqrt{1 - t^2}\ \ n_\perp(\phi).$$

Its midpoint is $m = Rt\, n(\phi)$, so the inverse is

$$C^{-1}(c) = \left(\operatorname{Arg}(m), \frac{\|\ m\ \|}{R}\right), \qquad m = \frac{z_+ + z_-}{2}.$$

Here Arg is chosen in $[0, 2\pi)$. The midpoint is nonzero because diameters were excluded. Hence $C\colon [0, 2\pi) \times (0, 1) \to \mathcal{C}_*$ is a bijection. Write $K = [0, 2\pi) \times (0, 1)$ and define the event collection explicitly by

$$\mathcal{F}_{\mathcal{C}_*} = \{B \subseteq \mathcal{C}_*\colon C^{-1}(B) \in \mathcal{B}(K)\}.$$

This makes C and its inverse measurable. The three chord-space triplets are

$$(\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_A), \qquad (\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_B), \qquad (\mathcal{C}_*, \mathcal{F}_{\mathcal{C}_*}, P_C).$$

Their sample space, event collection and long-chord event E are identical; their measures differ. The sampling maps below define these measures.

The length function L is **not** a bijection on chords: all orientations with the same t have the same length. The inverse image $L^{-1}(B)$ is still defined for every Borel set of lengths B.

A common source of uniform samples

Use the probability space

$$\Omega_0 = (0, 1) \times [0, 1), \qquad \mathcal{F}_0 = \mathcal{B}(\Omega_0), \qquad Q = \text{uniform area measure on } \Omega_0.$$

The coordinate random variables $U(u, v) = u$ and $V(u, v) = v$ are independent uniform variables. Endpoints of these intervals have measure zero. For each mechanism $j \in \{A, B, C\}$, define a sampling map

$$\Psi_j(u, v) = C\big(2\pi v, q_j(u)\big).$$

Each q_j given below is a continuous bijection of $(0,1)$. Since $C^{-1}\circ\Psi_j(u,v)=(2\pi v, q_j(u))$ is continuous into K, the definition of $\mathcal{F}_{\mathcal{C}_*}$ makes Ψ_j measurable. It is also a bijection onto $\mathcal{C}_*$. The event E is measurable because $C^{-1}(E)=[0,2\pi)\times(0,1/2)$.

The induced probability measure on chord events is

$$P_j(F)=Q\big(\Psi_j^{-1}(F)\big),\qquad F\in\mathcal{F}_{\mathcal{C}_*}.$$

The chord-length random variable on the uniform source space is $L_j=L\circ\Psi_j$. We now specify each q_j, its inverse and the distribution it produces.

A. Two independent uniform endpoints

The original procedure selects independent angles $A_1,A_2\sim\mathrm{Unif}[0,2\pi)$ and joins the points $R(\cos A_i,\sin A_i)$. The joint angle density is $1/(2\pi)^2$.

Let $\delta\in(-\pi,\pi]$ be the signed principal difference A_2-A_1 modulo 2π, and let $H=|\delta|/2$. In the deck's notation, $\alpha=|\delta|=2H$. The cases $\delta=0$ (coincident endpoints) and $|\delta|=\pi$ (diameters) have probability zero and are excluded for the chord chart. The map $(a_1,a_2)\mapsto(a_1,\delta)$ is a bijection apart from this null boundary and the angular wrap. For each a_1, the second coordinate is a translation modulo 2π, so the map preserves area. Hence (A_1,δ) has constant density $1/(4\pi^2)$ on the rectangle.

Now define $\Phi=(A_1+\delta/2)\bmod 2\pi$. The map $(a_1,\delta)\mapsto(\Phi,\delta)$ is also one-to-one on the chosen half-open angular ranges and preserves area: for fixed δ it is a translation in the first coordinate. Thus (Φ,δ) has the same constant density, which factors into uniform marginal densities. Applying $H=|\delta|/2$ gives the joint density

$$f_{\Phi,H}(\phi,h)=\frac{1}{2\pi}\,\frac{2}{\pi},\qquad 0\le\phi<2\pi,\quad 0<h<\pi/2.$$

Therefore Φ and H are independent, Φ is uniform, and $H\sim\mathrm{Unif}(0,\pi/2)$. The midpoint is in direction Φ, since the midpoint vector is proportional to $\cos(H)\,n(\Phi)$ and $\cos(H)>0$ almost surely.

The ordered-endpoint map is two-to-two because interchanging endpoints leaves the chord unchanged. Its inverse therefore recovers an **unordered pair**, not a unique ordered pair. From (ϕ,t), put $h=\arccos(t)$; the unordered endpoint angles are

$$\{(\phi-h)\bmod 2\pi,\ (\phi+h)\bmod 2\pi\}.$$

Thus an equivalent, bijective sampling parametrization of the unordered chords uses $U=2H/\pi$ and $V=\Phi/(2\pi)$:

$$q_A(u)=\cos(\pi u/2),\qquad \Psi_A^{-1}(c)=\left(\frac{2}{\pi}\arccos(T(c)),\frac{\mathrm{Arg}(m)}{2\pi}\right).$$

The length random variable and the inverse image of the event are

$$L_A(u,v)=2R\sin(\pi u/2),\qquad \Psi_A^{-1}(E)=\{(u,v)\colon u>2/3\}.$$

Consequently,

$$P_A(E)=Q(U>2/3)=1/3.$$

For $0<t<1$, the induced distribution of normalized midpoint distance is

$$F_{T,A}(t)=1-\frac{2}{\pi}\arccos(t)=\frac{2}{\pi}\arcsin(t),\qquad f_{T,A}(t)=\frac{2}{\pi\sqrt{1-t^2}}.$$

It is not uniform in radial distance.

B. Uniform radial distance

Choose $D\sim\mathrm{Unif}(0,R)$ and an independent direction $\Phi\sim\mathrm{Unif}[0,2\pi)$. Construct the perpendicular chord through $D\,n(\Phi)$. Equivalently,

$$q_B(u)=u,\qquad \Psi_B^{-1}(c)=\left(T(c),\frac{\mathrm{Arg}(m)}{2\pi}\right).$$

The original (D, Φ) joint density is $1/(2\pi R)$ on $(0, R) \times [0, 2\pi)$. In normalized coordinates, T has density $f_{T,B}(t) = 1$ and distribution function $F_{T,B}(t) = t$ for $0 < t < 1$.

$$L_B(u, v) = 2R\sqrt{1 - u^2}\ , \qquad \Psi_B^{-1}(E) = \{(u, v) : u < 1/2\}.$$

Therefore

$$P_B(E) = Q(U < 1/2) = 1/2.$$

C. Uniform midpoint in disk area

Choose a midpoint $M = (M_1, M_2)$ with Cartesian density $1/(\pi R^2)$ inside the disk. In polar coordinates $M = Rt\, n(\phi)$, the area element is $R^2 t\, dt\, d\phi$. Hence

$$f_{T,\Phi}(t, \phi) = \frac{t}{\pi} = 2t \cdot \frac{1}{2\pi}, \qquad 0 < t < 1.$$

Thus T and Φ are independent, Φ is uniform, and T has density $2t$ and distribution function $F_{T,C}(t) = t^2$ for $0 < t < 1$. A uniform draw U is transformed by

$$q_C(u) = \sqrt{u}\ , \qquad \Psi_C^{-1}(c) = \left(T(c)^2, \frac{\mathrm{Arg}(m)}{2\pi}\right).$$

The length random variable and event preimage are

$$L_C(u, v) = 2R\sqrt{1 - u}\ , \qquad \Psi_C^{-1}(E) = \{(u, v) : u < 1/4\}.$$

Therefore

$$P_C(E) = Q(U < 1/4) = 1/4.$$

The area ratio $\pi(R/2)^2/(\pi R^2)$ gives the same result.

Explicit comparison on the same coordinates

In every model, Φ is independent of T and uniform on $[0, 2\pi)$. The density with respect to $d\phi\, dt$ is $f_{T,j}(t)/(2\pi)$. Thus, for every $B \in \mathcal{F}_{\mathcal{C}_*}$,

$$P_j(B) = \iint_{C^{-1}(B)} \frac{f_{T,j}(t)}{2\pi}\, d\phi\, dt.$$

Equivalently, these densities define three measures on the same coordinate space $(K, \mathcal{B}(K))$.

Model	$T = q_j(U)$	**Inverse** $U = q_j^{-1}(T)$	**Density** $f_{T,j}(t)$, $0 < t < 1$	$P_j(E)$
Uniform endpoints	$\cos(\pi U/2)$	$(2/\pi)\arccos(T)$	$2/(\pi\sqrt{1 - t^2}\ )$	$1/3$
Uniform radial distance	U	T	1	$1/2$
Uniform midpoint area	$\sqrt{U}$	T^2	$2t$	$1/4$

For each row,

$$P_j(E) = \int_0^{1/2} f_{T,j}(t)\, dt = Q\big(\Psi_j^{-1}(E)\big).$$

There is no equality between $P_A(E)$, $P_B(E)$ and $P_C(E)$ to be contradicted: the measures are different. By contrast, on the common uniform source space $(\Omega_0, \mathcal{F}_0, Q)$, the measure Q stays fixed while the sampling maps change. The event preimages $\Psi_j^{-1}(E) = \{(u,v): L_j(u,v) > \sqrt{3}\ R\}$ are different sets, with Q-measures $1/3$, $1/2$, and $1/4$. Within a fixed model, a correct reparametrization preserves event probability.

A reparametrization within one fixed model

Retain Model B and set $W = T^2$. Since T is uniform on $(0,1)$,

$$P_B(W \le w) = P_B(T \le \sqrt{w}\ ) = \sqrt{w}\ , \qquad f_{W,B}(w) = \frac{1}{2\sqrt{w}}, \quad 0 < w < 1.$$

The inverse transformation is $T = \sqrt{W}$. The same event is now $W < 1/4$, and

$$P_B(E) = \int_0^{1/4} \frac{dw}{2\sqrt{w}} = \frac{1}{2}.$$

If W were instead declared uniform, its density would be changed to 1. That would give Model C and probability $1/4$. This would be a change of probability model, not merely a change of coordinates.

The distribution of chord length (additional calculation)

For $0 \le x \le 2R$, put $z = x/(2R)$. Because $L = 2R\sqrt{1-T^2}$ is decreasing in T,

$$P_j(L \le x) = P_j\left(\ T \ge \sqrt{1-z^2}\ \right).$$

The three chord-length distribution functions are

$$F_{L,A}(x) = \frac{2}{\pi}\arcsin(z), \qquad F_{L,B}(x) = 1 - \sqrt{1-z^2}\ , \qquad F_{L,C}(x) = z^2.$$

They are 0 for $x < 0$ and 1 for $x > 2R$. Substituting $x = \sqrt{3}\ R$ and taking $1 - F_{L,j}(x)$ gives $1/3, 1/2, 1/4$. These continuous laws assign zero probability to the equality at the threshold.

The set-based definitions follow Jagannathan, Chapters 1, 4 and 11. The comparison in common distance and angular coordinates is also developed in *Urban Operations Research*, §3.3.2. The explicit sampling maps and inverse calculations above are provided here in full.

4. The induced law is a probability measure

Let $X: (\Omega, \mathcal{F}, P) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be measurable and define $P_X(B) = P(X^{-1}(B))$. Nonnegativity follows from that of P, and

$$P_X(\mathbb{R}) = P(X^{-1}(\mathbb{R})) = P(\Omega) = 1.$$

If $B_1, B_2, \ldots$ are pairwise disjoint Borel sets, their inverse images are pairwise disjoint events, and inverse images preserve their union. Therefore

$$P_X\left(\bigcup_{n=1}^{\infty} B_n\right) = P\left(\bigcup_{n=1}^{\infty} X^{-1}(B_n)\right) = \sum_{n=1}^{\infty} P(X^{-1}(B_n)) = \sum_{n=1}^{\infty} P_X(B_n).$$

This proves that P_X is a probability measure. The same proof applies to the sampling maps Ψ_j and the induced measures P_j on the chord space. Injectivity is unnecessary for this construction: inverse images exist for all functions.

Distinct meanings of the notation f^{-1}

When $f\colon \mathsf{S} \to \mathsf{Y}$ is a bijection, $f^{-1}(y)$ denotes an element of S, the value of the inverse function at y. For any subset $B \subseteq \mathsf{Y}$, the notation $f^{-1}(B)$ denotes a subset of S, the inverse image. The latter is defined even when f is not one-to-one. In the Bertrand construction, C and each Ψ_j are bijections on the stated restricted domains, whereas the chord-length function L is not one-to-one.

5. Why a rotation-invariant probability cannot measure every subset

Section 1 uses Borel sets for the continuous model. Why not simply use every subset, as in the finite audit? Under the axiom of choice, the following construction shows that a normalized, countably additive, rotation-invariant probability on a circle cannot be defined on all its subsets.

The setting

Let $\Omega = [0, 1)$ and define addition modulo 1,

$$x \oplus y = \begin{cases} x + y, & x + y < 1, \\ x + y - 1, & x + y \geq 1, \end{cases}$$

with $\ominus$ the corresponding subtraction. Under $\oplus$ the set $[0, 1)$ is an abelian group; identifying 0 with 1 makes it a circle, and $x \mapsto x \oplus t$ is a rotation. For $A \subseteq [0, 1)$ write $A \oplus t = \{a \oplus t\colon a \in A\}$.

Suppose P assigns a nonnegative real number to **every** subset of $[0, 1)$ and satisfies

1. $P([0, 1)) = 1$;
2. countable additivity;
3. rotation invariance: $P(A \oplus t) = P(A)$ for all A and all $t \in [0, 1)$.

The construction

Let $\mathbb{Q}_0 = \mathbb{Q} \cap [0, 1)$, a countable subgroup of $([0, 1), \oplus)$. Define $x \sim y$ when $x \ominus y \in \mathbb{Q}_0$. This is an equivalence relation, and its classes are the cosets $x \oplus \mathbb{Q}_0$.

By the axiom of choice, let $V \subseteq [0, 1)$ contain exactly one element of each class.

Claim 1: the sets $V \oplus q$, for $q \in \mathbb{Q}_0$, are pairwise disjoint. Suppose $v_1 \oplus q_1 = v_2 \oplus q_2$ with $v_1, v_2 \in V$. Then $v_1 \ominus v_2 = q_2 \ominus q_1 \in \mathbb{Q}_0$, so $v_1 \sim v_2$. Since V has one element per class, $v_1 = v_2$, and therefore $q_1 = q_2$.

Claim 2: these sets cover $[0, 1)$. Given $x \in [0, 1)$, let $v \in V$ be the representative of the class of x. Then $q = x \ominus v \in \mathbb{Q}_0$ and $x = v \oplus q \in V \oplus q$.

So $[0, 1)$ is the union of countably many pairwise disjoint sets, each a rotation of V.

The contradiction

Write $c = P(V)$. Rotation invariance gives $P(V \oplus q) = c$ for every q, and countable additivity applied to the countable disjoint family of Claim 1, whose union is $[0, 1)$ by Claim 2, gives

$$1 = P([0, 1)) = \sum_{q \in \mathbb{Q}_0} P(V \oplus q) = \sum_{q \in \mathbb{Q}_0} c.$$

Since P is nonnegative, $c \geq 0$. If $c = 0$ the right-hand side is 0; if $c > 0$ the sum diverges. Neither equals 1, so no such P exists.

Moreover, if V were Lebesgue measurable, each $V \oplus q$ would be measurable with the same Lebesgue measure: addition modulo 1 is an ordinary translation on each of two pieces. The same countable sum would then contradict the Lebesgue measure of $[0, 1)$ being 1. Thus V is not Lebesgue measurable. This is the Vitali construction on the circle.

Scope of the conclusion

The contradiction uses both countable additivity and rotation invariance. Finite additivity alone does not justify the countably infinite sum above. Removing invariance changes the question: for example, the point mass $P(A) = \mathbf{1}_{\{0 \in A\}}$ is a countably additive probability on every subset of $[0, 1)$, but it is not rotation invariant.

The selection of V uses the axiom of choice; the argument does not give a membership rule for V. No restriction proved here prevents assigning probabilities to the Borel events used in this course. Intervals, disks, angular sectors and the sampling-map preimages in Section 3 are measurable. The construction explains a boundary of the all-subsets approach; it is not needed to carry out the practical calculations in the main lesson.

Advanced: a continuous ticket-closure model

The triplet is often not written out in routine probability calculations. To see what remains implicit, consider a ticket selected from a specified service process, and let T_c be its closure time, measured in hours. The question

> What is the probability that this ticket takes more than one day to close?

is written $P(T_c > 24)$, since one day is 24 hours. In an underlying ticket-system experiment, each outcome ω determines a closure time $T_c(\omega)$ in hours. The event of interest is the inverse image

$$T_c^{-1}((24, \infty)) = \{\omega \in \Omega : T_c(\omega) > 24\}.$$

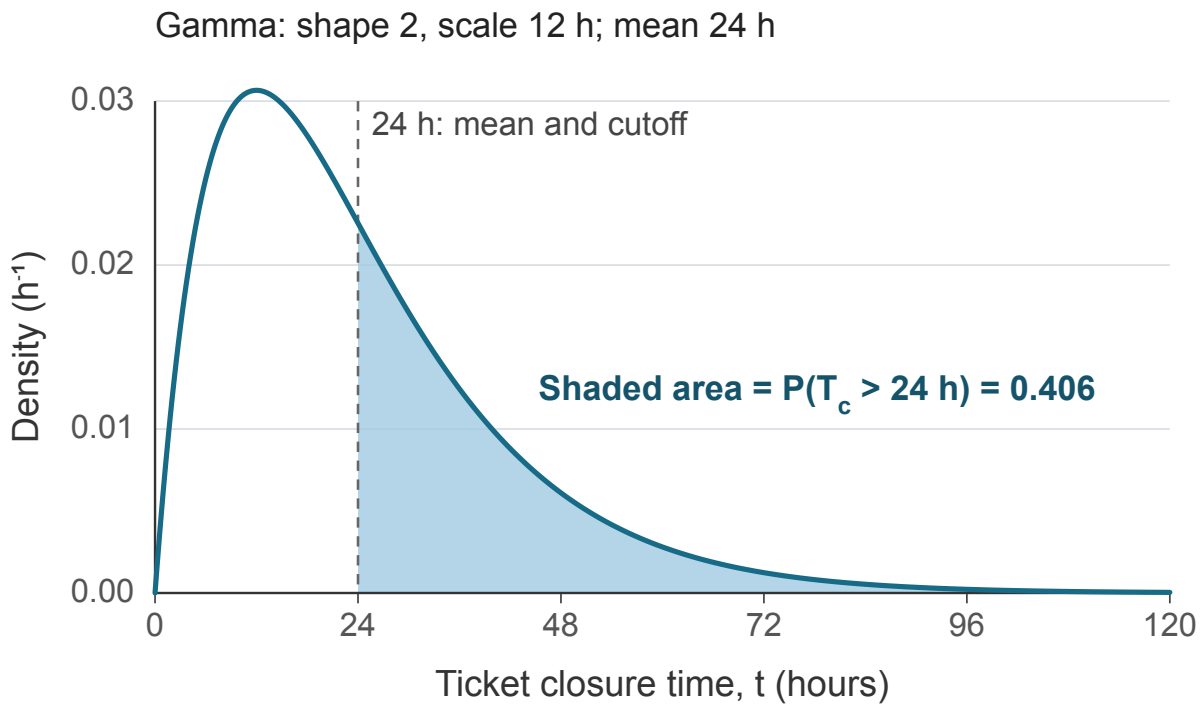
For an illustrative calculation, suppose the closure time follows a Gamma distribution with shape $\alpha = 2$ and scale $\theta = 12$ hours. These are hypothetical model parameters, not estimates from the six-ticket audit. Under the shape-scale parameterization, the density is

$$f_{T_c}(t) = \frac{t^{\alpha - 1} e^{-t/\theta}}{\Gamma(\alpha)\theta^{\alpha}}, \qquad t \geq 0.$$

Here $\alpha, \theta > 0$ and $\Gamma(a) = \int_0^{\infty} u^{a-1} e^{-u} du$ for $a > 0$; in particular, $\Gamma(2) = 1$. With $\alpha = 2$ and $\theta = 12$ hours, this becomes

$$f_{T_c}(t) = \frac{t}{12^2} e^{-t/12}, \qquad t \geq 0,$$

where t is measured in hours. Here α is the shape parameter, not the mean. The mean in this shape-scale parameterization is $E[T_c] = \alpha\theta = 24$ hours. The plotted curve shows this model and shades the area to the right of the 24-hour cutoff, which coincides with the mean.



Gamma density for ticket closure time T_c under the illustrative model. The vertical line marks both the mean and the 24-hour cutoff; the shaded right tail represents $P(T_c > 24\ \text{h})$.

For the six-ticket experiment in the main notes, probabilities were calculated by counting equally likely IDs: $P(C) = |C|/6$. A continuous ticket time cannot be handled by counting individual possible durations. Under a density model, the probability of an interval is the area under the density over that interval. A density value is not itself a probability; integration accumulates the density across the range. In the audit model, the corresponding finite calculation is a sum of singleton probabilities; for example, $P(C) = \sum_{t_i \in C} P(\{t_i\}) = 4/6$. The closure-time probability is the integral of its density over the event. With the substitution $u = t/12$,

$$\begin{aligned} P(T_c > 24) &= \int_{24}^{\infty} \frac{t}{144} e^{-t/12}\, dt \\ &= \int_{2}^{\infty} u e^{-u}\, du \\ &= \left[-(u+1)e^{-u}\right]_{2}^{\infty} \\ &= (2+1)e^{-2} = 3e^{-2} \approx 0.406. \end{aligned}$$

Thus, under this stipulated model, the probability that closure takes more than 24 hours is about 0.406. A different shape, scale or target ticket population would give a different value. Specifying the Gamma family alone would not suffice; the parameters and their parameterization are needed.

What a probability density specifies

The calculation above uses an area, not a density height. Formally, a random variable T has a **probability density function** f if f is nonnegative and measurable, $\int_{\mathbb{R}} f(t)\, dt = 1$, and

$$P(T \in D) = \int_{D} f(t)\, dt \quad \text{for every Borel set } D.$$

Here $f(t) = 0$ for $t < 0$. The same substitution checks normalization: $\int_0^{\infty} (t/144)e^{-t/12} dt = \int_0^{\infty} u e^{-u} du = 1$. A singleton has zero area, so $P(T_c = 24) = 0$, while a range can have positive probability. Not every random variable has a density; the Billing indicator has a discrete law.

The triplet for the closure-time model

For this calculation, use the following canonical probability space, which models closure time alone:

$(\Omega_c, \mathcal{F}_c, P_c)$.

In the calculation above, the shorthand $P(T_c > 24)$ denotes $P_c(T_c > 24)$ on this space. It is distinct from P, the uniform measure used for selecting one of the six audited ticket IDs.

1. **Sample space:** $\Omega_c = [0, \infty)$, the possible closure durations measured in hours.
2. **Event collection:** $\mathcal{F}_c = \mathcal{B}([0, \infty))$, the Borel subsets of the nonnegative real line: sets obtained by intersecting Borel subsets of $\mathbb{R}$ with $[0, \infty)$.
3. **Probability measure:** for every $E \in \mathcal{F}_c$,

$$P_c(E) = \int_E f_{T_c}(t)\,dt, \qquad f_{T_c}(t) = \frac{t}{144} e^{-t/12}, \quad t \geq 0.$$

The density integrates to one, so $P_c(\Omega_c) = 1$. Define the random variable by $T_c \colon \Omega_c \to [0, \infty)$, $T_c(t) = t$. Thus the event of closure after 24 hours is

$T_c^{-1}((24, \infty)) = (24, \infty) \in \mathcal{F}_c$,

and its probability is

$$P_c(T_c > 24) = P_c((24, \infty)) = \int_{24}^{\infty} f_{T_c}(t)\,dt.$$

This is one explicit probability space realizing the assumed Gamma law. A full ticket-system model could use outcomes that also include ticket type, priority and other information, with closure time as a function on that larger space. In routine calculations, $T_c \sim \text{Gamma}(\text{shape } 2, \text{scale } 12 \text{ h})$ states the induced law needed for probabilities about closure time without writing the whole triplet. The distribution of T_c does not by itself describe every other random quantity in the ticket system.

This is why many engineering and AI calculations work directly with random variables and their distributions after the probability-space foundation has been established.

6. Exercises on the supplement

1. Verify both $C^{-1}(C(\phi, t)) = (\phi, t)$ and $C(C^{-1}(c)) = c$ on the stated domains. Explain why the restriction on t is needed.
2. Verify both inverse identities for $q_A(u) = \cos(\pi u/2)$ on $(0, 1)$. Repeat for $q_C(u) = \sqrt{u}$.
3. Integrate each density in the comparison table over $(0, 1)$ and over $(0, 1/2)$.
4. Derive the three chord-length distribution functions from the distance distribution functions.
5. Prove that the inverse image of a union is the union of the inverse images. Use this to explain why the induced law is countably additive.
6. For one roll of a die, decide whether each of the following is a sample space, giving a reason: (a) $\{1, \ldots, 6\}$; (b) $\{\text{even}, \text{odd}\}$; (c) $\{\text{at most } 3, \text{at least } 3\}$; (d) $\{\text{less than } 3, \text{equal to } 3, \text{greater than } 3\}$.
7. In the Vitali construction, identify the step that fails if $\mathbb{Q}_0$ is replaced by a finite subgroup of $([0, 1), \oplus)$, and state what the argument then yields.
8. Connect the elementary experiments to Models A–C. (a) For Experiment I, find the second endpoint from $a = (1, 0)$ and angle θ. Then randomize the first endpoint uniformly, independently of θ, and prove that the two endpoint angles are independent and uniform. Show that the length law is unchanged. (b) Give Experiment II an independent uniform midpoint direction. Explain why its length law is unchanged. (c) Identify the relation between Experiment III and Model C. Finally, compare the probability of containing a in Experiment I and Model A. Does an equal length law imply an equal chord measure?

9. The density $f_{T,A}(t) = 2/(\pi\sqrt{1-t^2}\,)$ is unbounded as $t \uparrow 1$. Verify that it is integrable and find $P_A(T > 1-\varepsilon)$ for $0 < \varepsilon < 1$. What happens as $\varepsilon \downarrow 0$?
10. In Model B, write the event $T < 1/3$ in terms of $W = T^2$. Calculate its probability using the distribution function of W.